

2.2 THE BINARY NUMBER SYSTEM

The binary number system is a positional weighted system. The base or radix of this number system is 2. Hence, it has two independent symbols. The base itself cannot be a symbol. The symbols used are 0 and 1. A binary digit is called a bit. A binary number consists of a sequence of bits, each of which is either a 0 or a 1. The binary point separates the integer and fraction parts. Each digit (bit) carries a weight based on its position relative to the binary point. The weight of each bit position is one power of 2 greater than the weight of the position to its immediate right. The first bit to the left of the binary point has a weight of 2^0 and that column is called the *units column*. The second bit to the left has a weight of 2^1 and it is in the 2's column. The third bit to the left has a weight of 2^2 and it is in the 4's column, and so on. The first bit to the right of the binary point has a weight of 2^{-1} and it is said to be in the 1/2's column, the next right bit with a weight of 2^{-2} is in the 1/4's column, and so on.

The decimal value of the binary number is the sum of the products of all its bits multiplied by the weights of their respective positions. In general, a binary number with an integer part of $(n + 1)$ bits and a fraction part of k bits can be written as

$$d_n d_{n-1} d_{n-2} \dots d_1 d_0 . d_{-1} d_{-2} d_{-3} \dots d_{-k}$$

Its decimal equivalent is

$$(d_n \times 2^n) + (d_{n-1} \times 2^{n-1}) + \dots + (d_1 \times 2^1) + (d_0 \times 2^0) + (d_{-1} \times 2^{-1}) + (d_{-2} \times 2^{-2}) + \dots$$

In general, the decimal equivalent of the number $d_n d_{n-1} \dots d_1 d_0 . d_{-1} d_{-2} \dots$ in any number system with base b is given by

$$(d_n \times b^n) + (d_{n-1} \times b^{n-1}) + \dots + (d_1 \times b^1) + (d_0 \times b^0) + (d_{-1} \times b^{-1}) + (d_{-2} \times b^{-2}) + \dots$$

The binary number system is used in digital computers because the switching circuits used in these computers use two-state devices such as transistors, diodes, etc. A transistor can be OFF or ON, a switch can be OPEN or CLOSED, a diode can be OFF or ON, etc. These devices have to exist in one of the two possible states. So, these two states can be represented by the symbols 0 and 1, respectively.

2.2.1 Counting in Binary

Counting in binary is very much similar to decimal counting as shown in Table 2.1. Start counting with 0, the next count is 1. We have now exhausted all symbols; therefore we put a 1 in the column to the left and continue to get 10, 11. Thus, 11 is the maximum we can count using two bits. So, put a 1 in the next column to the left and continue counting; we can count 100, 101, 110, 111. The

2.2.2 Binary to Decimal Conversion

Binary numbers may be converted to their decimal equivalents by the positional weights method. In this method, each binary digit of the number is multiplied by its position weight and the product terms are added to obtain the decimal number.

EXAMPLE 2.5 Convert 10101_2 to decimal.

Solution

(Positional weights) 2^4 2^3 2^2 2^1 2^0

(Binary number) $1\ 0\ 1\ 0\ 1 = (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$
 $= 16 + 0 + 4 + 0 + 1$
 $= 21_{10}$

EXAMPLE 2.6 Convert 11011.101_2 to decimal.

Solution

$$\begin{array}{cccccccc} 2^4 & 2^3 & 2^2 & 2^1 & 2^0 & 2^{-1} & 2^{-2} & 2^{-3} \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \end{array} = (1 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0)$$

$$+ (1 \times 2^{-1}) + (0 \times 2^{-2}) + (1 \times 2^{-3})$$

$$= 16 + 8 + 0 + 2 + 1 + 0.5 + 0 + 0.125$$

$$= 27.625_{10}$$

EXAMPLE 2.7 Convert 1001011_2 to decimal.

Solution

An integer binary number can also be converted to an integer decimal number as follows. Starting with the extreme left bit, i.e. MSB, multiply this bit by 2 and add the product to the next bit to the right.

Multiply the result obtained in the previous step by 2 and add the product to the next bit to the right.

Continue this process as shown below till all the bits in the number are exhausted.

$$\begin{array}{cccccccc} 1 & & 0 & & 0 & & 1 & & 0 & & 1 & & 1 \\ & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & \\ & 1 \times 2 + 0 = 2 & & 2 \times 2 + 0 = 4 & & 4 \times 2 + 1 = 9 & & 9 \times 2 + 0 = 18 & & 18 \times 2 + 1 = 37 & & 37 \times 2 + 1 = 75 \end{array}$$

The above steps can be written down as follows:

Copy down the extreme left bit, i.e. MSB = 1	
Multiply by 2, add the next bit	$(1 \times 2) + 0 = 2$
Multiply by 2, add the next bit	$(2 \times 2) + 0 = 4$
Multiply by 2, add the next bit	$(4 \times 2) + 1 = 9$
Multiply by 2, add the next bit	$(9 \times 2) + 0 = 18$
Multiply by 2, add the next bit	$(18 \times 2) + 1 = 37$
Multiply by 2, add the next bit	$(37 \times 2) + 1 = 75$
The result is,	$1\ 0\ 0\ 1\ 0\ 1\ 1_2 = 75_{10}$

2.2.3 Decimal to Binary Conversion

There are two methods to convert a decimal number to a binary number. These are the reverse processes of the two methods used to convert a binary number to a decimal number.

First method: In this method, which is normally used only for small numbers, the values of various powers of 2 need to be remembered. For conversion of large numbers, we should have a

EXAMPLE 2.9 Convert 52_{10} to binary using the double-dabble method.

Solution

We divide the given decimal number successively by 2 and read the remainders upwards to get the equivalent binary number.

<i>Successive division</i>		<i>Remainder</i>	
2	52		
2	26		0
2	13		0
2	6		1
2	3	↑	0
2	1		1
	0		1

Reading the remainders from bottom to top, the result is $52_{10} = 110100_2$.

EXAMPLE 2.10 Convert 0.75_{10} to binary using the double-dabble method.

Solution

Multiply the given fraction by 2. Keep the integer in the product as it is and multiply the new fraction in the product by 2. Continue this process and read the integers in the products from top to bottom.

Given fraction	0.75
Multiply 0.75 by 2	1.50
Multiply 0.50 by 2	↓ 1.00

Reading the integers from top to bottom, $0.75_{10} = 0.11_2$.

EXAMPLE 2.11 Convert 105.15_{10} to binary.

Solution

Conversion of integer 105

Successive division	Remainder
2 105	
2 52	1
2 26	0
2 13	0
2 6	1
2 3	↑ 0
2 1	1
0	1

Reading the remainders from bottom to top, $105_{10} = 1101001_2$.

Conversion of fraction 0.15_{10}

Given fraction	0.15
Multiply 0.15 by 2	0.30
Multiply 0.30 by 2	0.60
Multiply 0.60 by 2	1.20
Multiply 0.20 by 2	↓ 0.40
Multiply 0.40 by 2	0.80
Multiply 0.80 by 2	1.60

This particular fraction can never be expressed exactly in binary. This process may be terminated after a few steps.

Reading the integers from top to bottom, $0.15_{10} = 0.001001_2$.

Therefore, the final result is, $105.15_{10} = 1101001.001001_2$.

2.2.4 Binary Addition

The rules for binary addition are the following:

$$0 + 0 = 0; \quad 0 + 1 = 1; \quad 1 + 0 = 1; \quad 1 + 1 = 10, \text{ i.e. } 0 \text{ with a carry of } 1.$$

EXAMPLE 2.12 Add the binary numbers 1101.101 and 111.011.

Solution

$$\begin{array}{rcccccccc}
 & 8 & 4 & 2 & 1 & 2^{-1} & 2^{-2} & 2^{-3} & \text{(Column numbers)} \\
 & 1 & 1 & 0 & 1 & \cdot & 1 & 0 & 1 \\
 + & & 1 & 1 & 1 & \cdot & 0 & 1 & 1 \\
 \hline
 & 1 & 0 & 1 & 0 & 1 & \cdot & 0 & 0 & 0
 \end{array}$$

In the 2^{-3} 's column $1 + 1 = 0$, with a carry of 1 to the 2^{-2} column
 In the 2^{-2} 's column $0 + 1 + 1 = 0$, with a carry of 1 to the 2^{-1} column
 In the 2^{-1} 's column $1 + 0 + 1 = 0$, with a carry of 1 to the 1's column
 In the 1's column $1 + 1 + 1 = 1$, with a carry of 1 to the 2's column
 In the 2's column $0 + 1 + 1 = 0$, with a carry of 1 to the 4's column
 In the 4's column $1 + 1 + 1 = 1$, with a carry of 1 to the 8's column
 In the 8's column $1 + 1 = 0$, with a carry of 1 to the 16's column

2.2.5 Binary Subtraction

The binary subtraction is performed in a manner similar to that in decimal subtraction. The rules for binary subtraction are:

$$0 - 0 = 0; \quad 1 - 1 = 0; \quad 1 - 0 = 1; \quad 0 - 1 = 1, \text{ with a borrow of } 1.$$

EXAMPLE 2.13 Subtract 111.111_2 from 1010.01_2 .

Solution

$$\begin{array}{rcccccccc}
 & 8 & 4 & 2 & 1 & 2^{-1} & 2^{-2} & 2^{-3} & \text{(Column numbers)} \\
 & 1 & 0 & 1 & 0 & \cdot & 0 & 1 & 0 \\
 - & & 1 & 1 & 1 & \cdot & 1 & 1 & 1 \\
 \hline
 & 0 & 0 & 1 & 0 & \cdot & 0 & 1 & 1
 \end{array}$$

In the 2^{-3} column, a 1 cannot be subtracted from a 0. So, borrow a 1 from the 2^{-2} column making the 2^{-2} column 0. The 1 borrowed from the 2^{-2} column becomes 10 in the 2^{-3} column. Therefore, in the 2^{-3} column, $10 - 1 = 1$.

In the 2^{-2} column, a 1 cannot be subtracted from a 0. So, borrow a 1 from the 2^{-1} column, but it is also a 0. So, borrow a 1 from the 1's column. That is also a 0, so borrow a 1 from the 2's column making the 2's column 0. This 1 borrowed from the 2's column becomes 10 in the 1's column. Keep one 1 in the 1's column, bring the other 1 to the 2^{-1} column, which becomes 10 in this column. Keep one 1 in the 2^{-1} column and bring the other 1 to the 2^{-2} column, which becomes 10 in this column. Therefore,

In the 2^{-2} 's column $10 - 1 = 1$
 In the 2^{-1} 's column $1 - 1 = 0$
 In the 1's column $1 - 1 = 0$

Now, in the 2's column, a 1 cannot be subtracted from a 0; so, borrow a 1 from the 4's column. But the 4's column has a 0. So, borrow a 1 from the 8's column, making the 8's column 0, and bring it to the 4's column. It becomes 10 in the 4's column. Keep one 1 in the 4's column and bring the second 1 to the 2's column making it 10 in the 2's column. Therefore,

$$\begin{array}{ll}\text{In the 2's column} & 10 - 1 = 1 \\ \text{In the 4's column} & 1 - 1 = 0 \\ \text{In the 8's column} & 0 - 0 = 0\end{array}$$

Hence, the result is 0010.011_2 .

2.2.6 Binary Multiplication

There are two methods of binary multiplication—the *paper* method and the *computer* method. Both the methods obey the following multiplication rules:

$$0 \times 0 = 0; \quad 1 \times 1 = 1; \quad 1 \times 0 = 0; \quad 0 \times 1 = 0$$

The paper method is similar to the multiplication of decimal numbers on paper. Multiply the multiplicand with each bit of the multiplier, and add the partial products. The partial product is the same as the multiplicand if the multiplier bit is a 1 and is zero if the multiplier bit is a 0.

EXAMPLE 2.14 Multiply 1101_2 by 110_2 .

Solution

The LSB of the multiplier is a 0. So, the first partial product is a 0. The next two bits of the multiplier are 1s. So, the next two partial products are equal to the multiplicand itself. The sum of the partial products gives the answer.

$$\begin{array}{r} 1101 \\ \times 110 \\ \hline 0000 \\ 1101 \\ 1101 \\ \hline 1001110 \end{array}$$

EXAMPLE 2.15 Multiply 1011.101_2 by 101.01_2 .

Solution

$$\begin{array}{r} 1011.101 \\ \times 101.01 \\ \hline 1011101 \\ 0000000 \\ 1011101 \\ 0000000 \\ 1011101 \\ \hline 11110100001 \end{array}$$

2.2.8 Binary Division

Like multiplication, division too can be performed by two methods—the paper method and the computer method. In the paper method, long-division procedures similar to those in decimal are used.

EXAMPLE 2.17 Divide 101101_2 by 110_2 .

Solution

Divisor 110 cannot go in the first three bits of the dividend, i.e. in 101. So, consider the first 4 bits 1011 of the dividend. 110 can go in 1011, one time with a remainder of 101. Next, 110 can go in 1010, one time with a remainder of 100. Next, 110 can go in 1001, one time with a remainder of 11. Finally, 110 can go in 110 with a remainder of 0.

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$$\begin{array}{r}
 110 \) \ 101101(111.1 \\
 \underline{110} \\
 1010 \\
 \underline{110} \\
 1001 \\
 \underline{110} \\
 110 \\
 \underline{110} \\
 000
 \end{array}$$

Therefore, $101101 \div 110 = 111.1$

EXAMPLE 2.18 Divide 110101.11_2 by 101_2 .

Solution

$$\begin{array}{r}
 101 \) \ 110101.11(1010.11 \\
 \underline{101} \\
 0110 \\
 \underline{101} \\
 111 \\
 \underline{101} \\
 101 \\
 \underline{101} \\
 000
 \end{array}$$

Therefore, $110101.11 \div 101 = 1010.11$